

STUDENT MANAGEMENT SYSTEM

Technical report

Talha Zia
22P-9228
BCS-6B

1. Project Overview

The Student Management System (SMS) is a professional web-based solution engineered to modernize academic administration. By digitizing student records and automating the grading lifecycle, the system reduces manual errors and provides instant access to academic data.

Core Functional Modules:

- **Student Profiling:** Comprehensive digital storage of student demographics and contact data.
- **Curriculum Management:** Tools to define courses and unique academic codes.
- **Dynamic Enrollment:** Bridge logic connecting students to specific academic terms.
- **Grade Processing:** Secure input for numeric marks and automated letter grade assignment.
- **Transcript Generation:** Dynamic report builder for individual student performance.

2. System Architecture

The application follows a standard 3-tier architecture utilizing the XAMPP stack:

- **Presentation Tier:** HTML5 and CSS3 with a persistent sidebar navigation UI.
- **Logic Tier:** PHP 8.x for session management, authentication, and CRUD logic.
- **Data Tier:** MySQL relational database ensuring data persistence and integrity.

3. Database Architecture

The database `student_management` utilizes 5 primary tables. The relational integrity is maintained via Foreign Key constraints.

3.1 Table: students (Core Profiles)

ATTRIBUTE	TYPE	DESCRIPTION
student_id	INT (PK)	Primary Key, Auto-increment.
first_name	VARCHAR	Student's given name.
last_name	VARCHAR	Student's family name.
email	VARCHAR	Contact email (Unique).
phone	VARCHAR	Contact number.
gender	VARCHAR	Gender identification.
dob	DATE	Date of Birth.
address	TEXT	Residential address.

3.2 Table: courses (Academic Catalog)

ATTRIBUTE	TYPE	DESCRIPTION
course_id	INT (PK)	Primary Key, Auto-increment.
course_name	VARCHAR	Title of the academic course.
course_code	VARCHAR	Unique identifier (e.g., CS101).

3.3 Table: enrollments (Relational Bridge)

ATTRIBUTE	TYPE	DESCRIPTION
enrollment_id	INT (PK)	Primary Key, Auto-increment.
student_id	INT (FK)	Links to students.student_id.
course_id	INT (FK)	Links to courses.course_id.
enroll_date	TIMESTAMP	Date of registration.

3.4 Table: grades (Results)

ATTRIBUTE	TYPE	DESCRIPTION
grade_id	INT (PK)	Primary Key, Auto-increment.
enrollment_id	INT (FK)	Links to enrollments.enrollment_id.
marks	INT	Numeric score (0-100).
grade	VARCHAR	Letter grade (A, B, C, etc.).

4. Technical Implementation & Security

4.1 Secure Authentication

Admin security is prioritized using MD5 encryption for password storage. Database queries utilize prepared statements to prevent SQL Injection.

```
$pass = md5($_POST['password']);  
$stmt = $conn->prepare("SELECT user_id FROM users WHERE username = ?  
AND password = ?");  
$stmt->bind_param("ss", $user, $pass);
```

4.2 Data Integrity Logic

The system prevents orphan records using PHP transactions. When a student record is deleted, the system automatically purges their associated grades and enrollments first to maintain database health.

5. Conclusion

This Student Management System serves as a robust platform for academic data management. By combining a clean user interface with a structured relational backend, it provides a scalable and secure tool for modern educational administration.